# §3. 27 lines on cubic surface

Convention: $X$ is always a smooth cubic surface / $k=\bar{k}$ $X \hookrightarrow \mathbb{P}^3$
char $k=0$

Goal: every smooth cubic surface in $\mathbb{P}^3$ contains exactly 27 lines.
every smooth cubic surface in $\mathbb{P}^3$ is isomorphic to $Bl_{P_1,\dots,P_6}\mathbb{P}^2$.

27.1.A. $X \hookrightarrow \mathbb{P}^3$ big $-K_X$. Consider $0 \to \mathcal{I}_X \to \mathcal{O}_{\mathbb{P}^3}|_X \to \mathcal{O}_X \to 0$

$\mathcal{O}_X(-\varphi) = \omega_X \otimes \mathcal{N}_X^\vee$ $\mathcal{I}_X = \mathcal{O}(-3)$ then $\mathcal{N}_X = \mathcal{O}_X(3)$

$\omega_X = \mathcal{O}_X(-1)$ i.e. $\mathcal{O}_X(1) = \mathcal{O}_X(-K_X)$. big $s_0, s_1, s_2, s_3 \in H^0(-K_X)$

27.1.B. $C \subset X$ smooth curve then $C \to \mathbb{P}^1 \xrightarrow{\text{line}} \mathbb{P}^3$ iff $(C^2)=-1$

consider $0 \to \mathcal{I}_C/\mathcal{I}_C^2 \to \Omega_X|_C \to \Omega_C \to 0$.

$\omega_X|_C = \omega_C \otimes \mathcal{O}_C(-C)$ take deg then

$\deg \mathcal{O}_C(-1) = (K_X \cdot C) = \deg K_C - (C \cdot C) = 2g-2-(C \cdot C)$

($\Leftarrow$) if $(C^2)=-1$ $2g-2+1 = \deg \mathcal{O}_C(-1) < 0$ then $g=0$ $C \simeq \mathbb{P}^1$
$\deg \mathcal{O}_C(-1) = -\deg C$ in $\mathbb{P}^3$

($\Rightarrow$) if $C \simeq \mathbb{P}^1$ $g=0$ $\deg \mathcal{O}_C(-1) = -2+(C \cdot C)$ $(C \cdot C)=-1$
$-\deg C = -1$ (line)

Caution: $C \subset \mathbb{P}^n$ line means $\deg C = 1$, i.e. $\deg \mathcal{O}_C(1)=1$. $C$ effective Cartier divisor

take $P, Q \in C$ and $H = \overline{PQ}$ then if $C \subset H$ $C \simeq \mathbb{P}^1$

if $C \subsetneq H$ $(C\cdot H) \geq 2$. but $(C\cdot H) = \deg C = 1$ contradiction.

but $C \hookrightarrow \mathbb{P}^1$ may not imply $\deg C = 1$ (not straight) $\mathbb{P}^1 \xrightarrow{O(2)} \mathbb{P}^3$

## 27.1.D. Moduli of cubic surfaces (may not smooth)

we use Hilbert scheme, codim 1 + hight polynomial $\Rightarrow$ principal $V_+(f)$. $\deg f = 3$.

consider $0 \to \mathcal{I}(-3) \to \mathcal{O} \to \mathcal{O}_X \to 0$

$$P_X(t) = \binom{3+t}{3} - \binom{t}{3} = \frac{(3+t)(2+t)(1+t)}{6} - \frac{t(t-1)(t-2)}{6}$$

$$= \frac{t^3+6t^2+11t+6 - t^3+3t^2-2t}{6} = \frac{3}{2}t^2 + \frac{3}{2}t + 6 = Q.$$

use this $t$ and consider $\mathrm{Hilb}^P_{\mathbb{P}^3} = \mathbb{P}^{19}$ by $\mathbb{P}^3 \xrightarrow{O(3)} \mathbb{P}^{19}$

① consider universal family $\mathcal{X} \subseteq \mathbb{P}^{19} \times \mathbb{P}^3$

$\downarrow$

$\mathbb{P}^{19}$

$F = \sum a_{ijke} x_0^i x_1^j x_2^k x_3^e = 0$ $p \in \mathrm{Sing}\, V_+(F)$ iff $\frac{\partial F}{\partial x_i}(p) = 0$ and $F(p) = 0$

Euler identity $\sum x_i \frac{\partial F}{\partial x_i} = 3F$ if $\mathrm{char} k \neq 3$

then consider $(a, x) \in \mathbb{P}^{19} \times \mathbb{P}^3$ $y := V\left(\frac{\partial F}{\partial x_i}(a, x)\right) \in \mathcal{X}$

② $\mathbb{P}^{19} \xleftarrow{h} Y \xrightarrow{\pi} \mathbb{P}^3$ consider fibre of $\pi$ WMA $p = [1,0,0,0]$

$\frac{\partial F}{\partial x_0} = 3a_{3000}$ $\frac{\partial F}{\partial x_1} = a_{2100}$ $\frac{\partial F}{\partial x_2} = a_{2010}$ $\frac{\partial F}{\partial x_3} = a_{2001}$.

then fibre over $p$ is intersection of hyperplanes $\mathbb{P}^{15}$

$Y$ is CM since $Y \hookrightarrow \mathbb{P}^{19} \times \mathbb{P}^3$ regular embedding (lci)

then by Miracle Flatness $Y/\mathbb{P}^3$ is $\mathbb{P}^{15}$-bundle. $\Rightarrow Y$ irreducible

then $\dim Y = 18$, calculate dimension of $\Delta = \mu(Y)$

27.1.F. $\mu$ is proper $\Rightarrow \mu(Y)$ closed. $\mu(Y)$ irreducible

first $\Delta \subseteq \mathbb{P}^{19}$ $\dim \Delta \leq 18$ consider $Y \to \mu(Y) = \Delta$

fibre of $X$ is $\mathrm{Sing}(X)$, then $\dim \mu^{-1}([X]) \geq \dim(\text{generic fibre}) = \dim Y - \dim \Delta$

(generic flatness + upper semicontinuity of dim of fibre)

Hope $\exists X$ (general) in $\mu(Y) = \Delta$ s.t. $\dim \mathrm{Sing}([X]) = 0$.

then $\dim \Delta = 18$ codim 1, then

Lemma. (27.1.E) $\exists$ cubic surface with single singular pt.

Consider $F = x_0(x_1^2 + x_2^2 + x_3^2) + x_1^3 + x_2^3 + x_3^3$

then $\Delta$ is prime Weil divisor in $\mathbb{P}^{19}$

(same proof works for deg $d$ hypersurface in $\mathbb{P}^n$)

27.2.A

$\mathcal{Z} \subseteq \mathbb{P}^{19} \times G(1,3)$ define data $\{(X, l) \mid l \subseteq X\}$.

$\mathcal{X} \subseteq \mathbb{P}^{19} \times \mathbb{P}^3$ $\mathcal{L} \subseteq G \times \mathbb{P}^3$ universal bundle

$\mathbb{P}^{19} = \mathrm{Hilb}^{Q}_{\mathbb{P}^3}$ $\mathcal{X}/\mathbb{P}^{19}$, then $\mathcal{Z} := \mathrm{Hilb}^{t+1}_{\mathcal{X}/\mathbb{P}^{19}}$ (discussed before poly $t+1 \Rightarrow$ line)

$\mathcal{Z}$ is flag Hilbert scheme i.e. $\mathcal{Z}(k) = \{(l, X) \mid l \subseteq X\}$ $\mathcal{Z}_{[X]} = \mathrm{Hilb}^{t+1}_{X/\mathbb{P}^{19}}$

$\mathcal{Z} = \mathrm{Hilb}^{Q, t+1}_{\mathbb{P}^3}$: $\mathrm{Sch}/k \to \mathrm{Set}$

$S \mapsto \{(\mathcal{L}, \mathcal{X}) \mid \mathcal{L} \subseteq \mathcal{X} \subseteq \mathbb{P}^3_S\}$

$= \{(\mathcal{L}, \mathcal{X}) \mid \mathcal{L}, \mathcal{X} \subseteq \mathbb{P}^3_S\}$

then forgetful morphism $\mathcal{Z} \to G$, $\mathcal{Z} \to \mathbb{P}^{19}$, $\mathcal{Z} \to G \times \mathbb{P}^{19}$

for $\mathcal{Z} \to G$ consider $0 \to T_{G/\mathbb{P}^{19}} \to T_{\mathcal{Z}} \to T_G \to \mathrm{Obs}_{\mathcal{L}}(X) = H^1(X, N_{X/\mathbb{P}^3} \otimes \mathcal{L}_{\mathcal{L}(3)})$

$T_{G/\mathbb{P}^{19}} = \mathrm{Def}_{\mathcal{L}}(X)$, $T_{\mathcal{Z}} = \mathrm{Def}(X, \mathcal{L})$, $T_G = \mathrm{Def}(\mathcal{L})$

$0 \to \mathcal{I}_\ell(3) \to \mathcal{O}(3) \to \mathcal{O}_\ell(3) \to 0$ then $H^0(\mathcal{O}_\ell(3)) \to H^1(\mathcal{I}_\ell(3)) \to H^1(\mathcal{O}_X(3))$

$0 \to \mathcal{I}_X(3) \to \mathcal{O}_{\mathbb{P}}(3) \to \mathcal{O}_X(3) \to 0$ $\quad H^1(\mathcal{O}_{\mathbb{P}}(3)) \to H^1(\mathcal{O}_X(3)) \to H^2(\mathcal{O}_{\mathbb{P}})$

$\mathcal{I}_{\mathbb{P}} \cong \mathcal{O}_{\mathbb{P}}$ $\quad 0 \quad 0 \quad 0$

$H^0(\mathbb{P}^3, \mathcal{O}(3)) \to H^0(H, \mathcal{O}(3)) \to H^0(\ell, \mathcal{O}_\ell(3))$ then $H^0(\mathcal{O}_\ell(3)) \to H^1(\mathcal{I}_\ell(3))$

$H^0(\mathbb{P}^3, \mathcal{O}(3)) \to H^0(X, \mathcal{O}_X(3)) \to H^0(\ell, \mathcal{O}_\ell(3))$ $\quad 20 \quad 4 \quad 0$

$T_\ell = H^0(\mathcal{I}_\ell(3)) = \ker(H^0(\mathcal{O}_X(3)) \to H^0(\mathcal{O}_\ell(3))$ of dim 15

consider $\mathcal{Z}$ fibre of $\mathcal{Z} \to G$ is $\{X \in \mathbb{P}^{19} \mid \ell \subset X\}$

$\mathbb{P}^{19}$ $\quad G = \mathrm{Grass}(4, 2)$ $\quad \dim_{\mathbb{C}} G$

Locally on $G$ $L$ is defined by $\begin{pmatrix} 1 & 0 & * \\ 0 & 1 & * \end{pmatrix}$ up to $\mathrm{GL}_4$

$\ell \subset X$ iff $a_{\text{some}} = 0$ ($4$ functions)

then locally $\mathcal{Z} = G \times \mathbb{P}^{15} \Rightarrow \mathcal{Z}$ is $\mathbb{P}^{15}$-bundle.

$\mathcal{Z}$ is smooth of dim 19. the multiplicity of $\ell \subset X$

is measured by $\mathcal{Z} \times_{\mathbb{P}^{19}} \mathrm{Spec}\, K[[X]]$

27.2.B now consider $\mathcal{Z} \xrightarrow{\pi} \mathbb{P}^{19}$.

first we have $\exists X \in \mathbb{P}^{19}$ $X$ only contains finite lines.

Part (27.1.C.) $x_0^3 + x_1^3 + x_2^3 + x_3^3 = 0$ contains exactly 27 lines

then upper semicontinuous $\Rightarrow \dim \pi^{-1}[X] \geq \dim(\mathcal{Z}) - \dim(\pi(\mathcal{Z}))$

$\dim \pi(\mathcal{Z}) = 19$ $\quad \pi(\mathcal{Z})$ closed irreducible then $\pi(\mathcal{Z}) = \mathbb{P}^{19}$

i.e. every cubic surface contains exactly 1 line.

and general cubic contains finite lines.

Next we calculate the multiplicity of $L \subset X$.

27.2.1. $\pi^{-1}(\mathbb{P}^{ps} \setminus D) \to \mathbb{P}^{ps} \setminus \Delta$ unramified.

$$\left( \begin{array}{l} \Leftrightarrow \forall x \in \mathbb{P}^{ps} \quad \pi^{-1}([X]) \text{ unramified} \\ \Leftrightarrow \forall X \quad \pi^{-1}([X]) \text{ etale}/k \\ \Leftrightarrow \forall X \quad \pi^{-1}([X]) = \coprod \operatorname{Spec} k \quad (k=\bar{k}) \end{array} \right)$$

proof it suffices to show

$$\forall \ell \in \pi^{-1}([X])(k) \qquad T_{\pi^{-1}([X])}(\ell) \text{ is dim } 0 \quad (=0)$$

$$\left( \begin{array}{l} \dim T_x Y \geq \operatorname{codim}_Y X \Rightarrow \dim \text{fibre} = 0 \quad \text{further } k[\varepsilon] \hookleftarrow A \text{ local Artin} \\ \text{then } m/m^2 = 0 \xrightarrow[\text{Nakayama}]{} m = 0 \quad \text{then } A = k \text{ since } k = \bar{k} \end{array} \right)$$

Deformation theory for relative deformation $L \subset X$

consider $\operatorname{Hilb}^P_X \quad P(t) = t+1 \quad (\deg 1 + \dim 1 \Rightarrow \forall Y \in \operatorname{Hilb}^P_X \; Y \text{ (irr)}$

If $\operatorname{Hilb}^P_{\mathcal{X}/\mathbb{P}^{ps}}$ Hilbert family, $\forall [X] \in \mathbb{P}^{ps}$ fibre of $\operatorname{Hilb}^P_{\mathcal{X}/\mathbb{P}^{ps}} / \mathbb{P}^{ps}$

$\parallel$ & $\quad$ is exactly $\operatorname{Hilb}^P_X$ = fibre of $\mathcal{Z}/\mathbb{P}^{ps}$

then $T_L \operatorname{Hilb}^P_X$ = {1st ord def of $L \subset X$} $= H^0(L, \mathcal{N}_{L/X})$ // CA.

$\parallel$

$\{k[\varepsilon] \to \operatorname{Hilb}^P_X\}$ = {flat family $\mathcal{I} \subset X \times \mathbb{C}$ $\mathcal{I}_0 = L$}

we calculate $\mathcal{N}_{L/X}$ by $0 \to \mathcal{N}_{L/X}^\vee \to \Omega_X|_L \to \Omega_L \to 0$

$\omega_X|_L = \omega_L \otimes \mathcal{N}_{L/X}^\vee$ since $\mathcal{O}_X(1) = \mathcal{O}_X(-K_X) \quad \omega_X = \mathcal{O}(-1)$

$\omega_L = \mathcal{O}(-2) \Rightarrow \mathcal{N}_{L/X}^\vee = \mathcal{O}(1) \quad \mathcal{N}_{L/X} = \mathcal{O}_L(-1)$ then $H^0(L, \mathcal{N}_{L/X}) = 0$

□

27.2.2. now $\mathcal{Z}' := \pi^{-1}(\mathbb{P}^1|\Delta) \to \mathbb{P}^1|\Delta$ unramified, miracle flatness $\Rightarrow$ flat

then $\mathcal{Z}'/(\mathbb{P}^1|\Delta)$ étale finite $\Rightarrow h^0(\mathcal{O}_{\pi^{-1}(L\cap\Delta)})$ constant $\Rightarrow$ #fibre $= 27$.

$\qquad\qquad\qquad

27.2.4.

then $\forall X_1$ lift $[X_1]$ to $W_1 \in W''$ then $W_1 \rightsquigarrow$ labeling $m_1 \cdots m_{27}$

$W_0 \rightsquigarrow \ell_1 \cdots \ell_{27}$

now $\mathcal{Z}' \subseteq M \times G$

$$\begin{array}{ccc} \mathcal{L} & \mathcal{L}_{\mathcal{Z}'} \to \mathcal{L} & \\ \downarrow & \downarrow \square \downarrow & \\ G & \mathcal{Z}' \to G & \end{array} \qquad \text{then} \quad \begin{array}{ccc} W' & \to & \mathcal{Z}' \\ & \searrow^{p_i} & \downarrow \\ & & \mathcal{Z} \end{array}$$

$(p_i^* \mathcal{L}_{\mathcal{Z}'} \cdot p_j^* \mathcal{L}_{\mathcal{Z}'})_p$ defined as $((p_i^* \mathcal{L}_{\mathcal{Z}'})(\phi) \cdot (p_j^* \mathcal{L}_{\mathcal{Z}'})(\phi))$ in $\mathbb{P}^7$

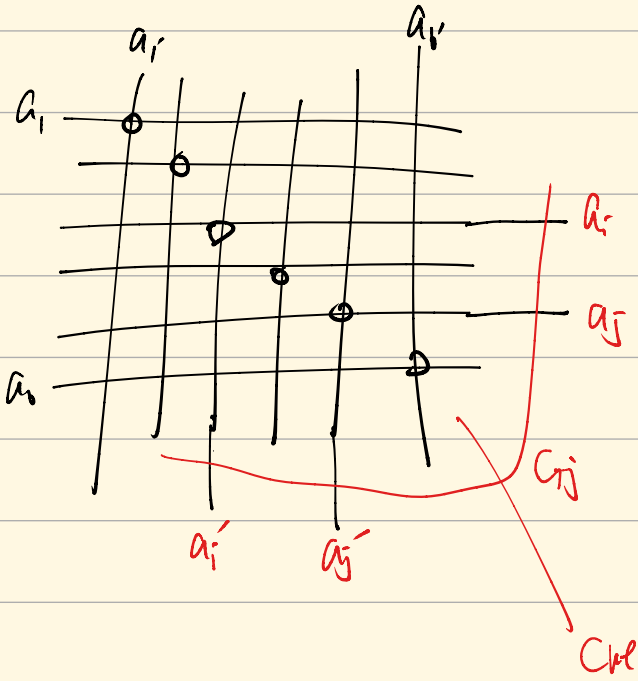
relative Carrier divisor

which is locally constant since $\chi$ of flat family is constant

$W''$ connected clopen subset $\Rightarrow \ell_i \cdot \ell_j = m_i \cdot m_j$

27.2.3/5 set $X_0 = (x_0^3 + x_1^3 + x_2^3 + x_3^3 = 0)$ then by calculation

the configuration is by $1, 1'$ $2, 2'$ $\cdots$ $6, 6'$ $a_i$ $a_i'$




| | | | | | |
|---|---|---|---|---|---|
| 12 | 13 | 14 | 15 | 16 | $C_{ij}$ |
| | 23 | 24 | 25 | 26 | |
| | | 34 | 35 | 36 | |
| | | | 45 | 46 | |
| | | | | 56 | |

(all of multiplicity 1)

Cut of $\{k, l\} \cap \{i, j\} = \emptyset$

take $a_i$ $a_i'$ then only $C_{12} \cdots C_{15}$ meeting both $C_{1i} \cap C_{ij} = \emptyset$

$\mathbb{P}^3 \setminus L \to \mathbb{P}^1$

27.3.A $X \subset \mathbb{P}^3$ WMA $V(x_0, x_1) \subseteq X$ consider $\mathbb{P}^3 \dashrightarrow \mathbb{P}^1$ fibre $\mathbb{A}^1$

$(x_0 : \cdots : x_3) \mapsto (x_0 : x_1)$

also graph

$B|_L \mathbb{P}^3 = V(x_1 T_0 - x_0 T_1) \subseteq \mathbb{P}^3 \times \mathbb{P}^1 \longrightarrow \mathbb{P}^1$ well-defined by projection

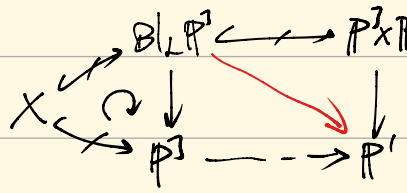
$(x_0 : \cdots : x_3) \times (T_0 : T_1)$

$(x_0 : \cdots : x_3) \mapsto (T_0 : T_1)$

but $B|_{L \cap X} X \hookrightarrow B|_L \mathbb{P}^3$ then

$\| $

$B|_L X = X$




Now consider $\ell \cap \ell' = \emptyset$ s.t. $\exists$ exactly 5 lines meeting both.

then $\ell' \hookrightarrow X \xrightarrow{\pi} \mathbb{P}^1$ gives isomorphism $\ell' \xrightarrow{\sim} \mathbb{P}^1$

Claim. $X \longrightarrow \mathbb{P}^1 \times \mathbb{P}^1$ is birational.

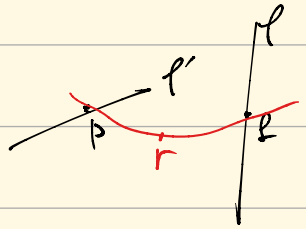
the inverse is $\mathbb{P}^1 \times \mathbb{P}^1 \longrightarrow \ell' \times \ell \longrightarrow X \subseteq \mathbb{P}^3$

$(p, q) \mapsto (\overline{pq} \cap X \setminus \langle p, q \rangle)$

first $\overline{pq} \subsetneq X$ for special $p, q$ (the five lines)

if $\operatorname{mult}_p(\overline{pq} \cap X) = 2$ $(p, q) \mapsto p$

similarly $\operatorname{mult}_q(\overline{pq} \cap X) = 2$. $(p, q) \mapsto q$.




Check birational: conversely $\forall$ r general pt, projection gives $(p, q)$

if $r \in \ell'$ $T_x r \cap \ell = q$ for

if $r \in \ell$ $T_x r \cap \ell' = p$ for

then isomorphism locus contains $X \setminus \ell_1 - \ell_5$

Remark: Fano's Variety of Lines.

27.3.B/27.3.1. $\rho(l_i)=p_i$ points.

$\rho: X \to \mathbb{P}^1\times\mathbb{P}^1$ is the blowup of $\mathbb{P}^1\times\mathbb{P}^1$ at five $p_i$

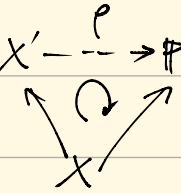
proof by projection $\rho(l_i)=(l_i\cap l', l_i\cap l)$

we use proof 1. by Castelnuovo's Criterion 28.6.1.

$\exists X \to X'$ s.t. $X = Bl_{p_i} X'$ $X'$ sm proj surface

$l_i \mapsto p_i'$

now $X' \dashrightarrow^{\rho} \mathbb{P}^1\times\mathbb{P}^1$ induces $X'\backslash\{p_1',\dots,p_5'\} \to X\backslash(p_1,\dots,p_5)$



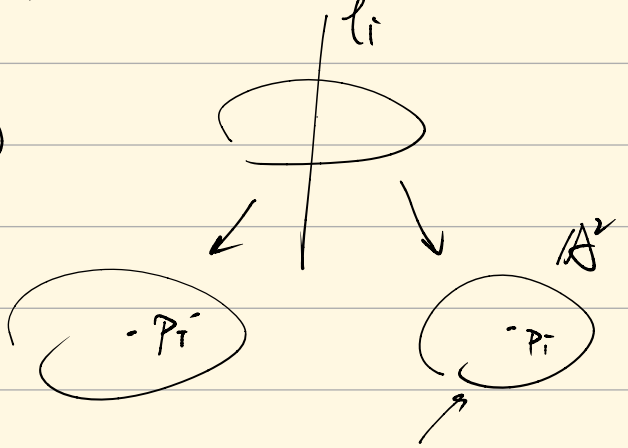
now consider $U = \mathbb{A}^1\times\mathbb{A}^1 \subseteq \mathbb{P}^1\times\mathbb{P}^1$

$\rho^{-1}(U) \to U$ $\rho-(p_1,p_2)\in \partial(\rho^{-1}(U))\times\partial(\rho^{-1}(U))$

Hartogs $\Rightarrow$ $\rho$ extends to $p_1'-p_5'$

and $\rho(p_i')=\rho(p_i)$

(local extension + gluing morphism)




shrink to affine open

s.t. $p_j\notin U$ $\forall j\neq i$

27.3.4. $Bl_p \mathbb{P}^1\times\mathbb{P}^1 = Bl_{p,q}\mathbb{P}^2$

proof $\mathbb{P}^2 \dashrightarrow \mathbb{P}^1\times\mathbb{P}^1$ fix $\underset{[0,0,1]}{l_1}\neq \underset{[0,1,0]}{l_2}\in\mathbb{P}^2$ $L=\overline{l_1 l_2}$

then $\pi_1 : \tilde{\mathbb{P}}^2 \to \mathbb{P}^1$ $x\mapsto \overline{\pi l_1}$ $\pi_2:\tilde{\mathbb{P}}^2\to\mathbb{P}^1$ ($\mathbb{P}^1$ seens as lines through $l_2$) $x\mapsto\overline{\pi p_2}$

$[x,y,z]\mapsto([x,y],[x,z])$

$\Phi:\mathbb{P}^2 \dashrightarrow \mathbb{P}^1\times\mathbb{P}^1$ indeterminent locus $\{l_1,l_2\}$

consider $Bl_{l_1,l_2}\mathbb{P}^2$ $\subseteq$ $\mathbb{P}^2\times\mathbb{P}^1\times\mathbb{P}^1$ $V_+(xT_2-yT_1, xS_2-zS_1)$

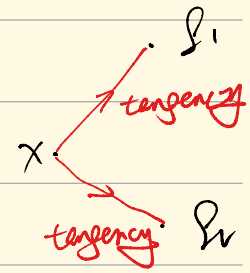
$x\,y\,z$ $T_1\,T_2$ $S_1\,S_2$

$\Phi$ lift as $\tilde{\Phi}: Bl_{l_1,l_2}\mathbb{P}^2 \xrightarrow{p_{23}} \mathbb{P}^1\times\mathbb{P}^1$ birational

$([x,y,z],[T_1,T_2],[S_1,S_2]) \mapsto ([T_1,T_2],[S_1,S_2])$

$\forall L_1, L_2 \in \mathbb{P}^1 \times \mathbb{P}^1$ if $L_1 \neq L_2$ $L_1 \cap L_2 = x$ (may be $\ell_1, \ell_2$)

then $(x, L_1, L_2) \longmapsto (L_1, L_2)$




if $L_1 = L_2 = L$ then $\forall x \in L$ $(x, L, L) \longmapsto (L, L)$

i.e. $\exists$ contracts $\Sigma$ and isomorphism outside $\Sigma$

$\varphi : B|_{\ell_1 \cup \ell_2} \mathbb{P}^2 \to \mathbb{P}^2$ then $\varphi^* L = \Sigma + E_1 + E_2$

$(\Sigma \cdot \Sigma) = (\varphi^* L - E_1 - E_2)^2 = (L \cdot \varphi_* \varphi^* L) - 2(L \cdot \varphi_* E_1) - 2(L \cdot \varphi_* E_2) - 1 - 1 = 4$

$\qquad\qquad\qquad\qquad\qquad\qquad\quad \| \qquad\qquad \| \qquad\qquad \|$

$\qquad\qquad\qquad\qquad\qquad\qquad\quad 1 \qquad\qquad 0 \qquad\qquad 0$

$\Sigma = B|_{p,q} L = L \Rightarrow \Sigma$ 有U-curve, 设S s.t. $B|_{\ell_1, \ell_2} \mathbb{P}^2$ 上 $\Sigma$, S 上 x

S smooth proj and $B|_{p,q} \mathbb{P}^2 = B|_x S$.

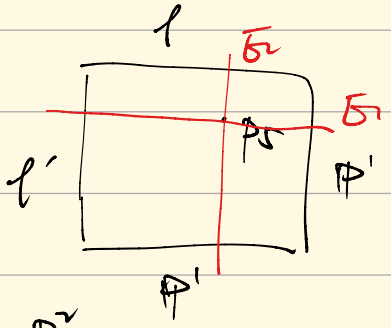
$S \to \mathbb{P}^1 \times \mathbb{P}^1$ same prove $\Rightarrow S \hookrightarrow \mathbb{P}^1 \times \mathbb{P}^1$ $S = B|_{(L_1, L_2)} \mathbb{P}^1 \times \mathbb{P}^1$ □

27.3.9. $X = B|_{p_1, \dots, p_5} \mathbb{P}^1 \times \mathbb{P}^1 = B|_{p_1, \dots, p_4} B|_{p_5} \mathbb{P}^1 \times \mathbb{P}^1 = B|_{p_1, \dots, p_4} B|_{\ell_1, \ell_2} \mathbb{P}^2$

$p_1, \dots, p_4 \in B|_{p_5} \mathbb{P}^1 \times \mathbb{P}^1 \setminus E = \mathbb{P}^1 \times \mathbb{P}^1 \setminus p_5 = B|_{\ell_1, \ell_2} \mathbb{P}^2 \setminus (E_1 \cup E_2)$

Claim: $p_1, \dots, p_4 \notin E_1 \cup E_2$

$p_1, \dots, p_5 \leftrightarrow \ell_1, \dots, \ell_5$




if $p_1 \in E_1$ $\ell_5 \cap \ell' = \ell_1 \cap \ell'$

$\ell_5 \cap \ell_1 \neq \emptyset$ contradiction

then $p_1 \notin E_1 \cup E_2$ $X = B|_{p_1, \dots, p_4, p_5, p_6} \mathbb{P}^2$

27.3.E. Conversely if $X = Bl_{p_1,\dots,p_6}\mathbb{P}^2$ $X \hookrightarrow \mathbb{P}^3$ cubic?

NO! if $p_1, \dots, p_6$ at special position, then $X$ not cubic surface.

(b) if $p_1, p_2, p_3, \dots, p_i$ collinear then $L := \overline{p_1 p_2 p_3}$ $(K_{\tilde{X}} \cdot \tilde{L}) = -\tilde{L}^2 + 2g - 2 = -(L^2) - 2$

$(L)^2 = (f^*L)^2 - i = 1 - i \le -2$ then $(K_X \cdot L) \ge 0$ contradicts to $-K_X$ ample.

(c) if $p_1, \dots, p_6 \subseteq C$ smooth conic, $g(C) = \frac{(d-1)(d-2)}{2} = 0$ $C \hookrightarrow \mathbb{P}^1$

consider $\tilde{C}$ $(K_{\tilde{X}} \cdot \tilde{C}) = -\tilde{C}^2 + 2g - 2 = -\tilde{C}^2 - 2$

$(\tilde{C})^2 = (f^*C - E_1 - \cdots - E_6)^2 = 0 - 1 - 1 - 1 - 1 - 1 - 1 = -2$ $(K_{\tilde{X}} \cdot \tilde{C}) = 0$ $\#$

27.3.6 $p_1, \dots, p_6 \in \mathbb{P}^2$ if $p_1, \dots, p_6$ satisfies { every 3 pts not collinear; every 6 pts not on a smooth conic.

then $X := Bl_{p_1,\dots,p_6}\mathbb{P}^2 \hookrightarrow \mathbb{P}^3$ as cubic.

proof: Fact: $Pic(Bl_{p_1,\dots,p_6}\mathbb{P}^2) = \mathbb{Z}H \oplus \mathbb{Z}E_1 \oplus \cdots \oplus \mathbb{Z}E_6$ $H = f^*h$, $h$ line in $\mathbb{P}^2$

then $K_X = aH + b_1E_1 + \cdots + b_6E_6$

$(K_X \cdot E_i) = -1 = -2 \Rightarrow (K_X \cdot E_i) = -1$ $b_i = 1$, WMA $p_1 \notin h$

$(K_X \cdot H)H = 2g - 2 = -2$ $(H \cdot H) = (h \cdot h) = 1$ $K_X \cdot H = -3$ then $a = -3$

$K_X = -3H + E_1 + \cdots + E_6$, first we need $-K_X$ very ample

$\forall$ effective irreducible curve $C \subseteq X$ $C = \tilde{C}$

if $C \cap E_i = \emptyset$ $(K_X \cdot C) = (-3H \cdot C) = -3(h \cdot f_*C) < 0$.

if $C \cap E_i \neq \emptyset$ 22.4.P $\Rightarrow f^*f_*C = C + \sum mult_{p_i}(f_*C) \cdot E_i$

$(K_X \cdot C) = (K_X \cdot f^*f_*C) + \sum mult_{p_i}(f_*C)$

$(f_*K_X \cdot f_*C) = -3 deg f_*C$

if $deg f_*C = 1$ $f_*C$ line $(K_X \cdot C) \le -3 + 2 = -1$

if $deg f_*C = 2$ $m_{p_i} + m_{p_j} \le \overline{p_i p_j} \cap f_*C = 2$ $(K_X \cdot C) < 0$

(irreducible curve is smooth)

$(K_X \cdot C) = 0$ iff $m_{p_i} = 1$ $\forall i$ & $\overline{p_i p_j} \cap f_* C = \{p_i, p_j\}$

Contradiction to $p_1, \dots, p_6$ not on a smooth conic.

WmA $\deg f_* C \geq 3$ take $C_1$ conic passing through $p_1, -\hat{p}_i, -p_6$

$(C_1 \cdot f_* C) = 2 \deg f_* C$ $\sum_{i \neq j} m_{p_i} \leq 2 \deg f_* C$ then $\sum m_{p_i} \leq 12 \deg f_* C$

$\sum m_i \leq 2 \cdot 6 \deg f_* C < 3 \deg f_* C$. $(K_X \cdot C) < 0$.

$(K_X)^2 = 9 - 6 = 3$ then $(-K_X)^2 = 3 > 0$. Nakai $\Rightarrow$ $K_X$ ample

Kodaira Vanishing: $H^i(X, -K_X) = H^i(X, K_X - 2K_X) = 0$ $\forall i > 0$. (Serre duality)

R-R: $h^0(X, -K_X) = \chi(X, -K_X) = \frac{(-K_X)(-2K_X)}{2} + \chi(\mathcal{O}_X) = \frac{6}{2} + 1 = 4$.

then $|-K_X| : X \dashrightarrow \mathbb{P}^3$ Fact: the morphism is $X \hookrightarrow \mathbb{P}^3$.